

Assignment 4 - LLM agents

Machine Learning S2026

GSM8k - The task

GSM8k - The Environment

GSM8k - Agent template

GSM8k - The task

Technical aspect of the assignment

enroll on the platform

starter code

Minimal score against the baseline

Using 🙌 Transformers library.

What is an LLM ?

A large language model (LLM) is an AI model trained on massive amounts of text data that can understand and generate human-like text, recognize patterns in language, and perform a wide variety of language tasks without task-specific training. They represent a significant advancement in the field of natural language processing (NLP).

LLMs characteristics

Scale: They contain millions, billions, or even hundreds of billions of parameters

General capabilities: They can perform multiple tasks without task-specific training

In-context learning: They can learn from examples provided in the prompt

Emergent abilities: As these models grow in size, they demonstrate capabilities that weren't explicitly programmed or anticipated

Limitations

Hallucinations: They can generate incorrect information confidently

Lack of true understanding: They lack true understanding of the world and operate purely on statistical patterns

Bias: They may reproduce biases present in their training data or inputs.

Context windows: They have limited context windows (though this is improving)

Computational resources: They require significant computational resources